Acid and Base Catalyzed Reactions of Active Methylene Compounds

Active Methylene Compounds - Overview

Active methylene compounds contain a methylene group (-CH_2-) flanked by one or more electron-withdrawing groups (EWG), making the hydrogen atoms on the methylene carbon relatively acidic. These compounds can be categorized as:

Mono-functional: Containing one electron-withdrawing group (e.g., ketones, aldehydes, esters)

Bi-functional: Containing two electron-withdrawing groups (e.g., β-dicarbonyl compounds, β-keto esters, malonates)

1. Acid Catalyzed Reactions of Mono-functional Active Methylene Compounds

Mono-functional active methylene compounds like ketones and aldehydes undergo various acid-catalyzed transformations through enol intermediates.

1.1 Keto-Enol Tautomerization

In acidic conditions, the carbonyl oxygen of ketones or aldehydes is protonated, increasing the electrophilicity of the carbonyl carbon and facilitating enolization:

R-CH_2-C=O\nR' → H^+ → R-CH_2-C^+-OH\nR' → -H^+ → R-CH=C-OH\nR'

Mechanism:

Protonation of the carbonyl oxygen by acid

Removal of α-hydrogen by a base (usually water in acidic conditions)

Formation of the enol tautomer

Example: Acid-catalyzed enolization of acetone:

CH_3-C=O-CH_3 → H^+ → CH_3-C^+-OH-CH_3 → -H^+ from α-carbon → CH_2=C-OH-CH_3

1.2 Aldol Condensation (Acid-Catalyzed)

Under acidic conditions, aldol condensation proceeds through enol intermediates rather than enolates:

R-CH_2-CHO

↓ H^+

R-CH_2-CH^+-OH

↓ -H^+

R-CH=CH-OH

↓ Attack on another\ncarbonyl

R-CH-CH_2-OH\n|\nCH\n|\nO\n|\nR'

↓ -H_2O

R-CH-CHO\n|\nCH=CHR'

Example: Acid-catalyzed aldol condensation of acetaldehyde:

2 CH_3CHO $\xrightarrow{H^+}$ $CH_3CH=CHCHO$ $\xrightarrow{H_2O}$ $CH_3CH_2CH_2CHO$

1.3 Halogenation (α-Halogenation)

Acid-catalyzed halogenation of ketones and aldehydes proceeds via enol intermediates:

R-CH_2-C=O-R' $\xrightarrow{H^+}$ R-CH=C-OH-R' $\xrightarrow{X_2}$ R-CHX-C=O-R'

Example: Bromination of acetone in acidic conditions:

CH_3COCH_3 $\xrightarrow{H^+}$ $CH_2=C(OH)CH_3$ $\xrightarrow{Br_2}$ $CH_2BrCOCH_3 + HBr$

1.4 Michael Addition

Acid-catalyzed Michael addition of mono-functional active methylene compounds to α,β-unsaturated carbonyl compounds:

R-CH_2-C=O-R' $\xrightarrow{H^+}$ R-CH=C-OH-R' → Unsupported markdown: list → R-CH_2-CH(R'')-C=O-R'

2. Acid Catalyzed Reactions of Bi-functional Active Methylene Compounds

Bi-functional active methylene compounds have enhanced acidity due to the presence of two electron-withdrawing groups, leading to more facile acid-catalyzed reactions.

2.1 Knoevenagel Condensation

Acid-catalyzed Knoevenagel condensation between a bi-functional active methylene compound and an aldehyde or ketone:

$CH_2(COOR)_2$ $\xrightarrow{H^+}$ CH(COOR)$_2$\n|\nH → Unsupported markdown: list → CH(COOR)$_2$\n|\nCH\n|\nR' $\xrightarrow{-H_2O}$ C=C\n/ Unsupported markdown: escape\nH(COOR)$_2$ R'

Example: Knoevenagel condensation of diethyl malonate with benzaldehyde:

$CH_2(COOEt)_2 + PhCHO$ $\xrightarrow{H^+}$ $Ph\text{-}CH=C(COOEt)_2 + H_2O$

2.2 Cyclization Reactions

Acid-catalyzed cyclization of 1,3-dicarbonyl compounds leading to heterocyclic ring formation:

$CH_3COCH_2COCH_3$ $\xrightarrow{H^+, NH_3}$ CH$_3$C=CH-C=CH$_3$\n| |\nO N\n Unsupported markdown: escape /\n H

Example: Formation of pyrazoles from 1,3-diketones:

$CH_3COCH_2COCH_3$ + H_2NNH_2 $\xrightarrow{H^+}$ CH$_3$C=CH-C=CH$_3$\n|| ||\nN-----N\n|\nH

2.3 Decarboxylation

Acid-catalyzed decarboxylation of β-keto acids and malonic acid derivatives:

$R\text{-}CH_2\text{-}CO\text{-}CH_2\text{-}COOH \xrightarrow{H^+,\ heat} R\text{-}CH_2\text{-}CO\text{-}CH_3 + CO_2$

Example: Decarboxylation of acetoacetic acid:

$CH_3COCH_2COOH \xrightarrow{H^+,\ heat} CH_3COCH_3 + CO_2$

3. Base Catalyzed Reactions of Mono-functional Active Methylene Compounds

Base-catalyzed reactions of mono-functional active methylene compounds proceed through enolate intermediates, which are stronger nucleophiles than enols.

3.1 Aldol Condensation

Base-catalyzed aldol condensation involves the nucleophilic addition of an enolate to another carbonyl compound:

$R\text{-}CH_2\text{-}CHO$

Base (OH^-)

↓

$R\text{-}CH^-\text{-}CHO$

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↓

R-CH-
CHO\n|\nCH\n|\nO\n|\nR'

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↓

R-CH=CH-CHO\n |\n R'

Example: Base-catalyzed aldol condensation of acetaldehyde:

2 CH_3CHO →NaOH→ $CH_3CH(OH)CH_2CHO$ →Unsupported markdown: list→ $CH_3CH{=}CHCHO$

3.2 Halogenation

Base-catalyzed halogenation of ketones and aldehydes proceeds via enolate intermediates:

$R\text{-}CH_2\text{-}CO\text{-}R'$ →Base→ $R\text{-}CH^-\text{-}CO\text{-}R'$ →X_2→ $R\text{-}CHX\text{-}CO\text{-}R' + X^-$

Example: Iodoform reaction (haloform reaction) of acetone:

CH_3COCH_3 →I_2, NaOH→ CH_3COCI_3 →NaOH→ $CH_3COO^-Na^+ + CHI_3$

3.3 Michael Addition

Base-catalyzed Michael addition involves the addition of an enolate to an α,β-unsaturated carbonyl compound:

$$R\text{-}CH_2\text{-}CO\text{-}R' \xrightarrow{\text{Base}} R\text{-}CH^-\text{-}CO\text{-}R' \xrightarrow{\text{Unsupported markdown: list}} R\text{-}CH_2\text{-}CH(CH_2\text{-}CO\text{-}R'')\text{-}CO\text{-}R'$$

Example: Michael addition of acetone to methyl vinyl ketone:

$$CH_3COCH_3 + CH_2{=}CHCOCH_3 \xrightarrow{\text{NaOH}} CH_3COCH_2CH_2CH_2COCH_3$$

4. Base Catalyzed Reactions of Bi-functional Active Methylene Compounds

Bi-functional active methylene compounds undergo a variety of base-catalyzed reactions due to their enhanced acidity and the stability of the resulting carbanions.

4.1 Alkylation

Base-catalyzed alkylation of bi-functional active methylene compounds with alkyl halides:

$$CH_2(COOR)_2 \xrightarrow{\text{NaOEt}} CH^-(COOR)_2 \xrightarrow{\text{R'-X}} R'\text{-}CH(COOR)_2$$

Example: Malonic ester synthesis:

$$CH_2(COOEt)_2 \xrightarrow{\text{Unsupported markdown: list}} CH_3CH(COOEt)_2 \xrightarrow{\text{Unsupported markdown: list}} CH_3CH(CH_3)(COOEt)_2 \xrightarrow{\text{Unsupported markdown: list}} CH_3CH(CH_3)COOH$$

4.2 Claisen Condensation

Base-catalyzed Claisen condensation between two ester molecules:

$$2\ CH_3COOEt \xrightarrow{\text{NaOEt}} CH_3COCH_2COOEt + EtOH$$

Mechanism:

Base removes a proton from the α-carbon of one ester molecule

The resulting enolate attacks the carbonyl carbon of another ester

Elimination of alkoxide results in the β-keto ester

4.3 Dieckmann Condensation

Base-catalyzed intramolecular Claisen condensation of diesters:

$EtOOC-(CH_2)_n-COOEt$ NaOEt ▸ $O=C-(CH_2)_{n-2}-CH-COOEt$\n |

Example: Dieckmann condensation of diethyl adipate:

$EtOOC-(CH_2)_4-COOEt$ NaOEt ▸ $O=C-(CH_2)_3-CH-COOEt$\n |

4.4 Knoevenagel Condensation

Base-catalyzed Knoevenagel condensation between an active methylene compound and an aldehyde or ketone:

$CH_2(CN)_2$ Base ▸ $CH^-(CN)_2$ Unsupported markdown: list ▸ $R-CH(OH)-CH(CN)_2$ Unsupported markdown: list ▸ $R-CH=C(CN)_2$

Example: Knoevenagel condensation of malononitrile with benzaldehyde:

$CH_2(CN)_2 + PhCHO$ Piperidine ▸ $Ph-CH=C(CN)_2 + H_2O$

4.5 Michael Addition

Base-catalyzed Michael addition of bi-functional active methylene compounds to α,β-unsaturated carbonyl compounds:

$CH_2(CN)_2$ Base ▸ $CH^-(CN)_2$ Unsupported markdown: list ▸ $NC-CH_2-CH_2-CH(R)-CN$

Example: Michael addition of ethyl acetoacetate to methyl vinyl ketone:

CH_3COCH_2COOEt + CH_2=$CHCOCH_3$ $\xrightarrow{\text{NaOEt}}$ $CH_3COCH(CH_2CH_2COCH_3)COOEt$

4.6 Robinson Annulation

Base-catalyzed Robinson annulation involves a Michael addition followed by an intramolecular aldol condensation:

Cyclohexanone + CH_2=$CHCOCH_3$

Base

↓

Michael Adduct

Intramolecular\nAldol

↓

2-Methyl-2-cyclohexenone

Example: Robinson annulation in the synthesis of 2-methyl-2-cyclohexenone:

O=C\n/ Unsupported markdown: escape\n$CH_2$ CH_2\n| |\n$CH_2$ CH_2\nUnsupported markdown: escape /\n$CH_2$	Unsupported markdown: list →	O=C\n/ Unsupported markdown: escape\n$CH_2$ C\n| ||\n$CH_2$ C\nUnsupported markdown: escape /\n$CH_2$ CH_3

5. Comparison of Acid and Base Catalyzed Reactions

Aspect	Acid Catalyzed	Base Catalyzed
Active Species	Enol	Enolate
Nucleophilicity	Moderate	High
Reaction Rate	Generally slower	Generally faster
Selectivity	Often less selective	Often more selective
Reversibility	Typically more reversible	Less reversible
Side Reactions	More common	Less common with controlled conditions

6. Summary

Active methylene compounds are versatile building blocks in organic synthesis, capable of undergoing numerous transformations under both acidic and basic conditions:

Mono-functional compounds (ketones, aldehydes) react through enol (acid) or enolate (base) intermediates in reactions like aldol condensation and halogenation.

Bi-functional compounds (β-dicarbonyl compounds, malonates) have enhanced reactivity due to increased acidity and stabilized carbanions, participating in reactions like alkylation, Claisen condensation, and Michael addition.

Understanding the reaction mechanisms and conditions for these transformations enables chemists to selectively synthesize complex organic molecules from relatively simple starting materials, making active methylene chemistry a cornerstone of organic synthesis.